

# CoRoT-2b

R-1908

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_CoRoT-2\_260630 · created 2026-08-05T23:26:46+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2461219.9836 +/- 0.0036 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.114 +/- 0.064                 |
| Transit depth (Rp/Rs)^2                           | 1.29 +/- 1.46 [%]               |
| Orbital Inclination (inc)                         | 89.23 +/- 2.02                  |
| Airmass coefficient 1 (a1)                        | 2.6408916 +/- 5e-06             |
| Airmass coefficient 2 (a2)                        | -0.46 +/- 0.6                   |
| Scatter in the residuals of the lightcurve fit is | 0.00017 %                       |
| Best Comparison Star                              | #1 - [608, 214]                 |
| Optimal Aperture                                  | 20.5                            |
| Optimal Annulus                                   | 26.67                           |
| Transit Duration (day)                            | 0.00416 +/- 0.00015             |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                        |
|--------------------------------------|--------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.114 $\pm$ 0.064 vs 0.1667 (deviation 0.82 $\sigma$ ) |
| Duration fit vs published            | 0.00416 d vs 0.095 d (deviation 605.6 $\sigma$ )       |
| Mid-time O-C                         | -2508.3 min                                            |
| Depth SNR                            | 1.8                                                    |
| Residual scatter                     | 0.00017 %                                              |

### Light curve

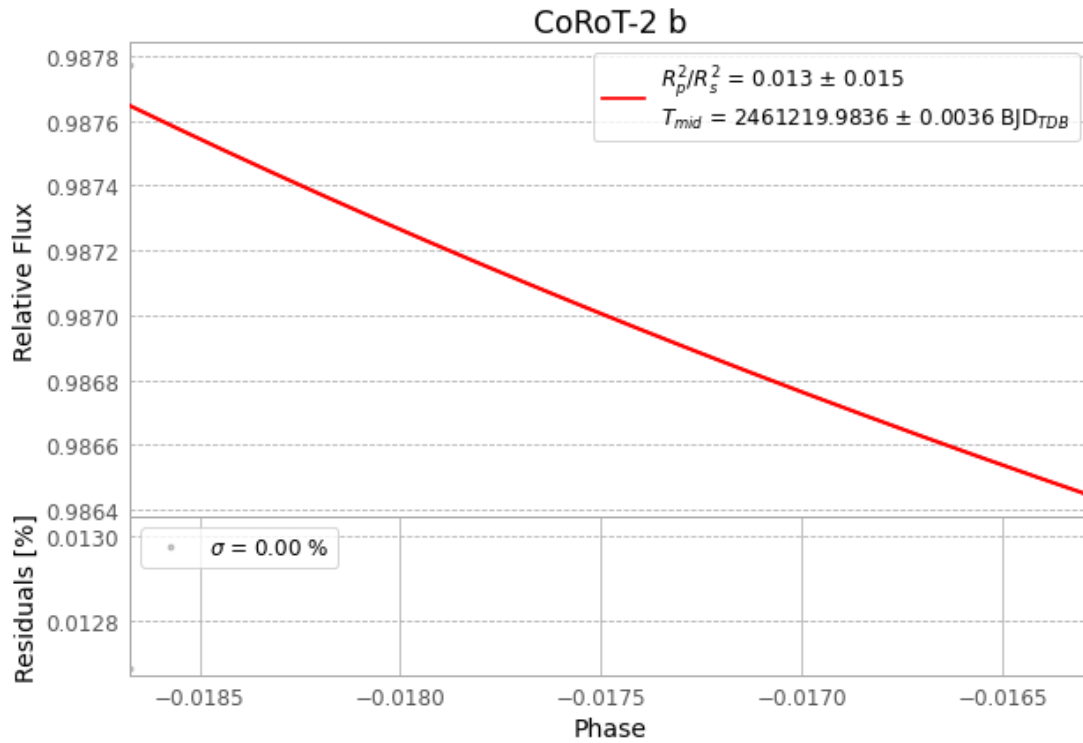


Figure 1 — FinalLightCurve\_CoRoT-2 b\_2026-06-29.png (SHA-256 642e54e77da04472...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [267, 251], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 91a798dfea33f77d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

141 FITS frames (63 MB), CoRoT-2260630032805.FITS.gz → CoRoT-2260630072716.FITS.gz. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-260630.log (16a1fdbba986...), FinalLightCurve\_CoRoT-2 b\_2026-06-29.png (642e54e77da0...), FinalLightCurve\_CoRoT-2 b\_2026-06-29.csv (0741f8471b32...)

## Compute resources

Wall time 222.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-05 23:26Z.